# Fabric defect detection: a Roboflow and PyTorch comparison

*Single-label classification of fabric images into Defect free, Defected, and Waste. Both models are ResNet-18, trained on the same 70/20/10 split.*

## 1. Dataset

The images were labeled in Roboflow and frozen as dataset version 1 (Poorva Fabric defect detection, IDM 2), exported in folder format on 29 October 2026. The export contains 4,995 source images: 3,499 train, 999 validation, and 497 test, which is the requested 70/20/10 split. Every image was auto-oriented and stretched to 224x224. Two augmentations were applied to the training images only: a horizontal flip with probability 50%, and a brightness change between -15% and +15%. Roboflow kept two copies of each training image, so the training folder has 6,998 files.

The classes are uneven. The training folder holds 6,304 Defected, 670 Defect free, and 24 Waste images. The test folder holds 449 Defected, 47 Defect free, and 1 Waste image. A model that almost always predicts Defected will score well on overall accuracy. Per-class recall is the number that shows whether Defect free and Waste are actually recognized.

## 2. The two models

The Roboflow model is sanya-mohanty/poorva-fabric-defect-detection-idm-2-1-resnet18-t1. It is a ResNet-18 started from the ImageNet checkpoint, trained with Roboflow's default hyperparameters. Inference in this benchmark calls the hosted API at serverless.roboflow.com. On the test split, Roboflow's evaluation reports precision 94.5%, recall 91.7%, and F1 93.0% at an operating threshold of 61% confidence.

The PyTorch model is also a ResNet-18, loaded from ImageNet weights with the final layer replaced by three outputs, then fine-tuned on the exported train and validation folders. Training used AdamW with learning rate 0.0003, weight decay 0.0001, batch size 32, and a cosine learning-rate schedule for 15 epochs. The training loader applied the same two augmentations, random horizontal flip and brightness jitter of 15%. Validation and test images were only resized and normalized. The run used the Apple GPU (MPS), took 19.2 minutes (1154 seconds), and reached its best validation accuracy of 97.0% at epoch 14.

Both models are scored on one shared list of 50 test images, saved in eval_images.txt so the comparison cannot drift. The list has 25 Defect free, 24 Defected, and 1 Waste image. It contains every Waste test image and about half of the Defect free test images, so it is a harder sample than the full test set. predictions.csv holds the Roboflow output for these 50 images: file name, predicted label, and confidence.

## 3. Accuracy, training time, and latency

| Measure | Roboflow ResNet-18 | PyTorch ResNet-18 |
|---|---|---|
| Test precision | 94.5% | 92.7% macro |
| Test recall | 91.7% | 90.2% macro |
| Test F1 | 93.0% | 91.4% macro |
| Test accuracy | see note below | 95.8% (476/497) |
| Accuracy on the same 50 images | 82% (41/50) | 78% (39/50) |
| Training time | from the training email | 19.2 min |
| Mean latency on the 50 | 382 ms | 11.3 ms |
| Median latency on the 50 | 365 ms | 11.2 ms |

Read the two accuracy columns on the same footing. Roboflow's 94.5 / 91.7 / 93.0 figures come from its model evaluation on the test split, taken at the 61% threshold. PyTorch has no reject option: every image gets the class with the highest score, and the precision, recall, and F1 above are macro averages, so each class counts equally. Under that rule the PyTorch model scores 95.8% accuracy on all 497 test images. Most of that score is the Defected class, which it recalls at 98.2% (441 of 449). Defect free recall is 72.3% (34 of 47). The single Waste test image is correct, and one image is not evidence.

On the shared 50 images, Roboflow is correct on 41 and PyTorch on 39. The four-point gap is two images. Both percentages sit well below the full-test accuracy because this sample is half Defect free, and Defect free is the class both models miss.

Latency is where the two setups differ. Roboflow's mean of 382 ms per image (median 365 ms, 95th percentile 515 ms) includes sending the file to the hosted API and waiting for the response. PyTorch's mean of 11.3 ms is measured on the same machine and includes loading the image and preprocessing; the forward pass alone averages 8.8 ms. The hosted number is the cost of calling the API. The local number is the cost of running the network beside the camera.

PyTorch training took 19.2 minutes for 15 epochs, about 77 seconds per epoch, on a laptop GPU. Roboflow does not write its training duration into the exported metrics. The duration is on the training job, or in the completion email, and that value should be entered in the table above before the report is submitted. The hosted job used Roboflow's default recipe. The local job used the learning rate, batch size, and epoch count in section 2.

## 4. Failure analysis

Of the 50 shared images, 11 are wrong in at least one model, and every one of them is a Defect free image predicted as Defected. Nine are wrong in both models, most of them with confidence above 0.97. Two are wrong only in PyTorch. On defect-free-137, Roboflow scores Defect free at 0.616 and PyTorch scores Defected at 0.899. On defect-free-54, Roboflow scores Defect free at 0.638 and PyTorch scores Defected at 0.613. Those two Roboflow decisions sit just above the 61% threshold, so they are shallow wins.

The full PyTorch test set shows the same boundary. It makes 21 errors: 13 Defect free images called Defected, and 8 Defected images called Defect free. Waste is never confused with either class. The PyTorch confusion matrix is:

| True \ predicted | Defect free | Defected | Waste |
|---|---|---|---|
| Defect free | 34 | 13 | 0 |
| Defected | 8 | 441 | 0 |
| Waste | 0 | 0 | 1 |

The missed images are hard to tell apart by eye once they have been stretched to 224x224 and compressed. Many of the files are only 2 to 5 KB. A plain measure of texture, the standard deviation of the grayscale pixels, lines up with the mistakes. A typical Defect free test image has a standard deviation near 3. The Defected images called Defect free are the smoothest Defected images, down to 2.2. The Defect free images called Defected include the most textured Defect free images, with a standard deviation near 16, close to the Defected median of 11.8. Both networks are separating swatches by how much the weave varies, and on these edge cases that variation does not match the label. Because the wrong answer often comes with confidence above 0.99, raising the threshold does not remove them. The Waste test image is a fold rather than a stain, and both models label it Waste. With 12 Waste images in the original training split and one in the test split, that single correct call is not a result to lean on.

## 5. Conclusion

The two ResNet-18 trainings land in the same place on accuracy. Roboflow's published test F1 is 93.0%, and the PyTorch macro F1 is 91.4%. On the 50 shared images Roboflow is two images ahead, 82% against 78%. The local model answers in about 11 ms, against about 382 ms for the hosted API, and it trained in 19.2 minutes on a laptop. The errors are concentrated on one boundary: smooth Defected swatches and textured Defect free swatches. More images in Defect free and Waste, and a pass over the labels where those two classes look alike, would change the result more than a larger network would.